Single-electron transistor

In all parts of this problem, the elementary charge $e = 1.6 \times 10^{-19}\ C$ may appear in the answer.

The single-electron transistor is an electronic component whose operation is based on the *Coulomb blockade* phenomenon observed at low temperatures (on the order of several kelvin). In this problem, we will investigate this effect and explain the operating principle of a single-electron transistor at zero temperature.

Note: In electrical circuits, the "ground" refers to the point where the potential is defined as zero. The "ground" symbol is shown in the figure.

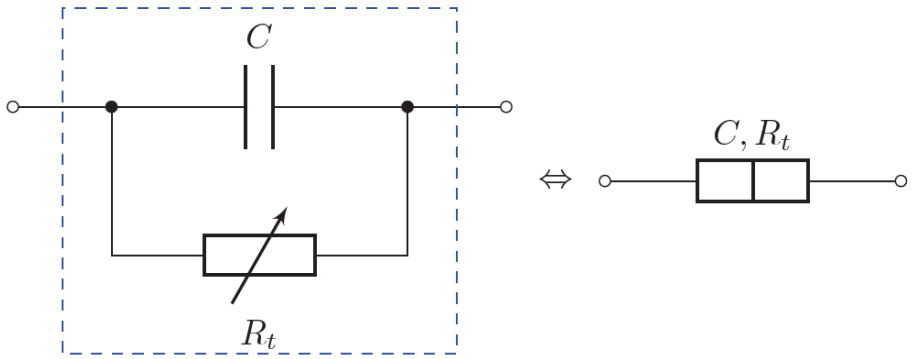


Part A. Tunneling effect (3.0 points)

The Coulomb blockade phenomenon is fundamentally based on *tunneling*, or *the tunnel effect*. Its essence lies in the fact that, under certain conditions, a system can overcome potential energy barriers. Tunneling is possible if the final energy does not exceed the initial energy. The excess energy releases as heat.

Tunneling constitutes the sequential transfer of individual elementary charges between the plates of a tunnel junction. In the "classical" model of this system, electron transport occurs through a resistor R_t connected in parallel with a capacitor. Energy dissipates on the resistor. The resistance remains finite when tunneling is energetically permissible, and becomes infinite otherwise.

Further, the charge or energy of the tunnel junction shall imply the charge or energy of the capacitor.



Tunnel junction circuit (left) and its symbol in electrical circuit diagrams (right)

Let us consider a tunnel junction disconnected from the power supply.

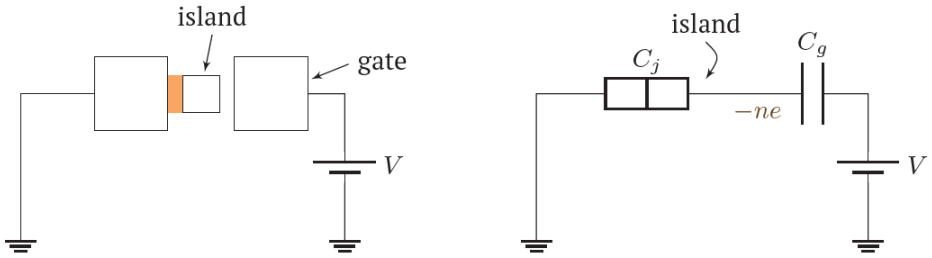
A.1	Write the expression for the energy of a capacitor with capacitance C (in the equivalent circuit of a tunnel junction) when its plates hold a charge equal to n electron charges. Now let the value n increase by one. What is the change in the capacitor's energy ΔW?	0.1 pt

Hereafter, we consider tunnel junctions connected in an electrical circuit with other elements. In circuit diagrams, a tunnel junction is represented as two touching rectangles.

Single-electron box

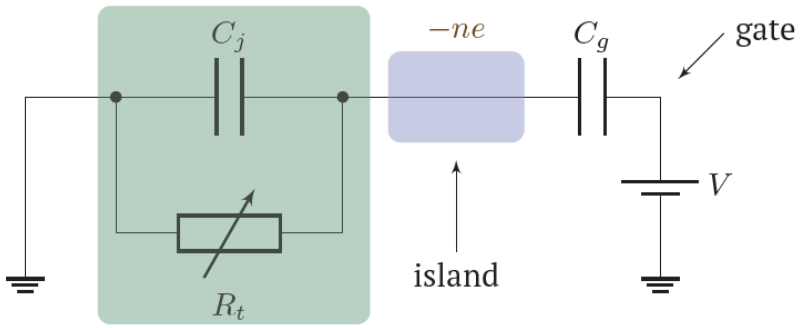
The single-electron box is a system comprising three aligned electrodes. The central electrode is called the island, while the lateral electrodes are called the banks. The island contains n electrons — a quantity that may vary through tunneling events. Note that n can be **either positive or negative** (indicating an excess or deficiency of electrons on the island).

Between the island and the left bank lies a tunnel junction with capacitance C_j, while the island and right bank form a conventional capacitor with capacitance C_g. In this configuration, the right bank is designated as the *gate*. A constant potential difference V is maintained between the banks. The corresponding electrical circuit diagram, specifying all component parameters, is shown in the figure.



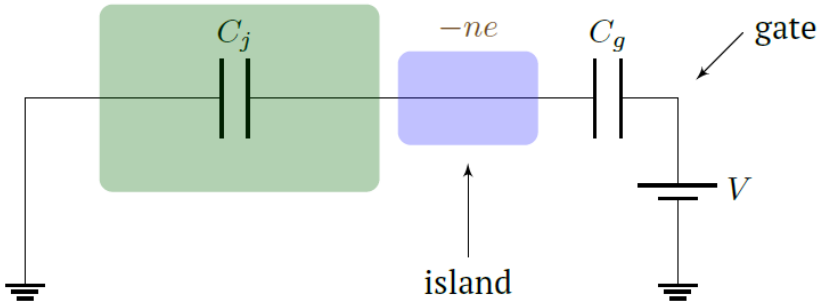
Single-electron box

The "classical" model of tunneling when it is energetically permissible.



Tunneling is permissible

The "classical" model of tunneling when it is energetically forbidden.



Forbidden tunneling

A.2	Derive expressions for the charges on the capacitor and tunnel junction q_g and q_j. The capacitances of the capacitor and tunnel junction are C_g and C_j, respectively, the source voltage is V, and the number of electrons on the island is n.	0.3 pt

A.3	Derive the expression for the change in energy of the system consisting of the capacitor and tunnel junction when the number n of electrons on the island increases by one. Express the answer in terms of n, C_g, $C = C_j + C_g$.	0.3 pt

A.4	Derive expressions for the work done by the voltage source A and the heat dissipation Q in the circuit during the tunneling of one electron onto the island (corresponding to an increase by one in the number n of island electrons). Express the answers in terms of n, C_g, C, V.	0.5 pt

Hereafter, we will consider tunneling to be allowed when the amount of heat Q dissipated in the circuit is greater than zero.

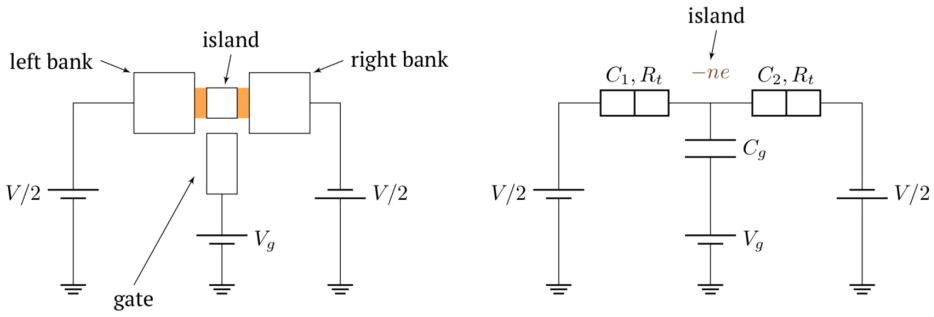
A.5	Assume that the heat Q is dissipated during current flow through the tunnel resistance R_t. Derive the expression for the current I_t through R_t during tunneling of one electron onto the island. Also, specify the constraint on n for which this expression is valid (i.e., when tunneling is energetically permissible). Express the answers in terms of n, C_g, C, V, R_t.	0.5 pt

A.6	Assuming tunneling is energetically permissible, express the current I flowing through the source in terms of the tunnel current I_t. The positive current direction is toward the negative terminal of the source. The expression may also include the quantities C_g, C, V. Consider processes corresponding to both an increase and a decrease in the number of electrons on the island.	0.3 pt

A.7	Plot the $I(V)$ characteristics for $n = 0$ and $n = 1$ on the single graph. Display the dependencies over a sufficiently wide voltage range to capture all distinctive features.	1.0 pt

Part B. Single-electron transistor (4.5 points)

The single-electron transistor consists of an island connected to two banks via tunnel junctions and a gate that forms a capacitor with the island. The gate is purely capacitively coupled (its resistance is infinite). A voltage $+V/2$ is applied to the left bank, and a voltage $-V/2$ is applied to the right bank. The transistor's circuit diagram and equivalent circuit, with specified parameters of the electronic components, are shown in the figure.



Single-electron transistor.

B.1	Let the island contain n electrons. Express the island's potential φ in terms of n, V, V_g, C_1, C_2, C_g.	0.2 pt

B.2	Derive the expression for the total work done by the voltage sources during the tunneling of one electron onto the island from the left (first) bank. Express your answer in terms of V, V_g, C_1, C_2, C_g.	1.0 pt

B.3	Let the island contain n electrons. Express the total heat dissipated in the circuit during the tunneling of one electron onto the island from the left (first) bank. Express the answer in terms of n, V, V_g, C_1, C_2, C_g.	1.0 pt

B.4	Derive the condition (inequality) for V corresponding to heat dissipation positivity during the tunneling from the left bank **onto** the island. Express the answer in terms of n, V, V_g, C_g, C_2.	0.3 pt

B.5	Derive conditions for V analogous to those in question B4 in cases of tunneling **onto** the island from the right bank and tunneling **from** the island onto the both banks. Sketch a diagram of regions corresponding to the existence of n electrons on the island after sufficient time has elapsed. The diagram should be plotted in axes V vs V_g. For each region specify the values of n on the diagram.	1.5 pt

Consider regions on the (V, V_g) diagram where, after sufficient time, only two transitions are energetically permissible: $n \to n+1$ through one junction and $n+1 \to n$ through the other junction. In this case, a steady current flows through the island from one bank to the other. We now determine its magnitude.

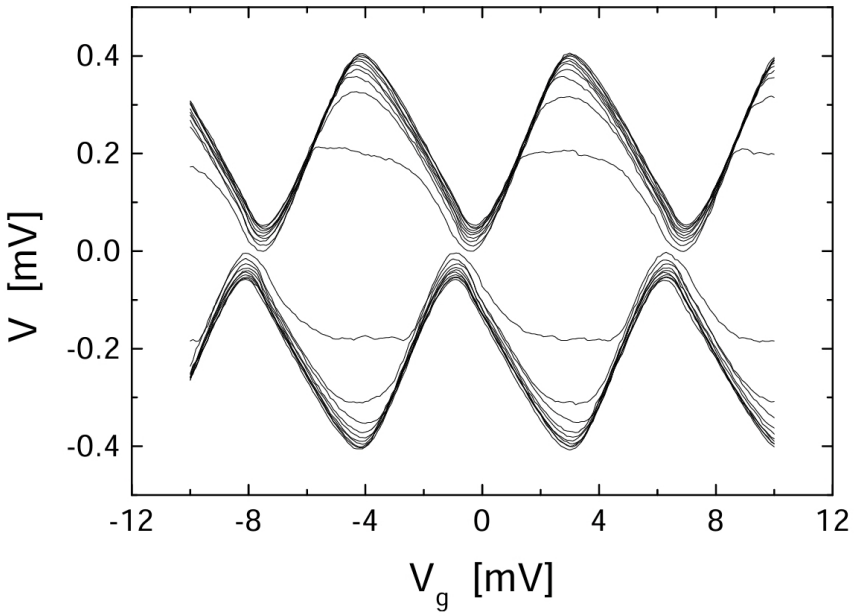
Since the island's charge remains unchanged after a pair of tunneling events, the charges on the capaci-

tors also remain constant. Consequently, the resulting current is determined by the sum of the tunneling times through the first and second junctions.

B.6	Assume the tunnel junctions are identical: each has a capacitance C_1 and tunnel resistance R_t. Consider voltage pairs (V_g, V) for which, after sufficient time, only states with $n = 0$ and $n = 1$ electrons on the island are possible. Determine the magnitude of the tunnel current $I(V_g, V)$ in the circuit. The positive current direction is from the left bank to the right bank.	0.5 pt

Part C. Tunneling at finite temperature (2.5 points)

C.1	The graph shows the experimental $V(V_g)$ dependence at low temperature $T = 10\ \mathrm{mK}$, measured for fixed current values $I \approx 1\ \mathrm{pA}$. Based on the results obtained in Part B, determine the capacitances $C_1 = C_2$ and C_g.	0.5 pt



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Consider a fixed positive temperature T. Let $\Gamma^{(i)}_{n\to n\pm1}$ denote the probability **per unit time** (rate) of tunneling to/from the island through the i-th junction, and let $p(n)$ be the probability of finding n electrons on the island.

Note: If a process has a rate Γ, then during time dt it occurs with probability $dp = \Gamma dt$.

C.2	Let $\Gamma_{n\to n\pm1} = \Gamma^{(1)}_{n\to n\pm1} + \Gamma^{(2)}_{n\to n\pm1}$. An electron can tunnel to/from either bank. Express the time derivative $dp(n)/dt$ of the probability of having n electrons on the island in terms of the probabilities $p(n)$, $p(n \pm 1)$ and the rates $\Gamma_{n\to n\pm1}$, $\Gamma_{n\pm1\to n}$.	0.3 pt

We will examine the steady-state current flow through the transistor (meaning all probabilities $p(n)$ are constant). We also assume there exists a critical number n_0 of electrons on the island, beyond which tunneling ceases: $p(n) = 0$ and $\Gamma_{n-1\to n} = 0$ for $n > n_0$.

C.3	From the stated assumptions and established boundary condition, derive a relation connecting $p(n)$ and $p(n+1)$ in the form $p(n+1) = A \cdot p(n)$, where A is an expression containing only the rates $\Gamma_{m \to m \pm 1}$ for various m.	0.5 pt

We will consider low temperatures ($kT \ll e^2/C$) and small voltages: $|V| \ll e/C$, $|V_g - e/2C_g| \ll e/C_g$. According to quantum mechanics, the general form for the tunneling rate per unit time to/from the island through the i-th junction is:

$$\Gamma^{(i)}_{n \to n \pm 1} = \frac{1}{e^2 R_t} \frac{Q^{(i)}_{\pm}(n)}{1 - \exp\left[-\dfrac{Q^{(i)}_{\pm}(n)}{kT}\right]}$$

where $Q^{(i)}_{\pm}(n)$ is the heat dissipated during the tunneling process.

C.4	Using the expression derived in question B3 and analogous relations, show that $\Gamma_{n \to n+1}$ for $n > 0$ and $\Gamma_{n \to n-1}$ for $n < 1$ **decay** exponentially with changing n. Assume that $T \to 0$.	0.3 pt

Hereafter, all $p(n)$ except $p(0)$ and $p(1)$ will be treated as zero.

C.5	Express the probabilities $p(0)$ and $p(1)$ in terms of the rates $\Gamma_{0 \to 1}$ and $\Gamma_{1 \to 0}$. Take into account that $p(0) + p(1) = 1$.	0.5 pt

C.6	Write the expression for the current through the transistor. Express the answer in terms of $\Gamma^{(i)}_{0 \to 1}$, $\Gamma^{(i)}_{1 \to 0}$.	0.4 pt

Substituting the expressions for heat into the formula for the tunneling rates $\Gamma^{(i)}_{n \to n \pm 1}$ yields, after extensive calculations, the current's dependence on voltage and temperature. This pronounced dependence enables the use of a single-electron transistor as a low-temperature measurement device, as explored in the science festival problem.